Jesse Curran

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Education

University of California, San Diego **Expected Mar. 2027**
M.S. Computer Science, Robotics *La Jolla, CA*

- Introduction to Robotics, Artificial Intelligence, Robot Sensing & Estimation, Human–Robot Interaction. In progress: Math for Robotics, Embedded Systems, Research Project.

University of Southern Denmark **Aug. 2026**
International Elite Summer School in Robotics & Entrepreneurship (5 ECTS) *Odense, Denmark*

- Programmed pick-and-place tasks on a Universal Robots UR3e arm.

Oregon State University **May 2024**
B.S. Computer Science *Corvallis, OR*

Research Experience

Graduate Researcher, UC San Diego *La Jolla, CA*
Autonomous Vehicles Lab | Advisor: Henrik I. Christensen **Jan. 2026 – Present**

- Developing on-road and off-road air–ground autonomy with LiDAR–semantic traversability mapping, Starlink teleoperation, and risk-aware navigation (IEEE IV 2027 submission in preparation).
- Developed CPU-only target re-identification for RGB-D person following on Spot (Kalman tracking, re-ID memory, ridge classifier); in offline replay, raised identity F-score from 40.0 to 69.3 under sun–shade–indoor lighting at 10.5 FPS (contributing to a paper in preparation).
- Built and field-tested a ROS 2 Nav2 navigation stack with costmaps for a GEM e6 autonomous vehicle, with live RViz monitoring and on-robot debugging; tuned basic control and drove it autonomously for 0.5 mi.
- Built camera–LiDAR perception: YOLO 2D detection on TensorRT, a C++ LiDAR 3D detection pipeline (ground segmentation, DBSCAN clustering, bounding boxes), and LiDAR–RealSense depth fusion.
- Integrated FAST-LIO2 LiDAR–inertial localization and measured its error over a 1-mile driving route.
- Ran PX4 UAV waypoint-following flights alongside the autonomous vehicle in joint air–ground field tests; wrote a C++ OpenCV ROS 2 detection node for one UAV.

Professional Experience

Robotics Autonomy Engineer Intern **Oct. 2026 – Present**
Naval Information Warfare Center Pacific | Secret Clearance (Interim) *San Diego, CA*

- Researching and developing drone-swarm and UAV autonomy for RESTORE Lab, supporting UAV and robotic platform integration.

Senior Financial Analyst **Sept. 2021 – June 2026**
Providence Health & Services *Renton, WA*

- Built and deployed an automated access system supporting 5,000+ users, improving operating efficiency tenfold.
- Led the development of a productivity analytics platform that replaced a $10M vendor solution; built a hospital financial forecasting model that achieved 3% MAPE.
- Trained four analysts in programming, automation, data modeling, and financial analysis.

Selected Robotics Projects

State Estimation and SLAM | *ICP LiDAR SLAM with loop closure; visual–inertial EKF on $SE(3)$* **2026**

Technical Skills

Languages:	Python, C++, C
Robotics Software:	ROS 2, Nav2, PX4, GTSAM, FAST-LIO2, RViz, Foxglove
Perception & ML:	PyTorch, OpenCV, TensorRT, OpenVINO, YOLO
Methods:	SLAM, mapping, state estimation, sensor fusion, calibration, path planning, PID control
Hardware:	LiDAR, Intel RealSense RGB-D cameras, IMUs; Spot, PX4 UAVs, GEM e6, UR3e
Dev Tools:	Linux, Git, Docker (on-robot deployment)